

COMP 1101 - Final

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1. Which of the following for loops prints all the multiples of 4 from 4 to 48.

a)

```
for i = 4 : 48
    disp(i);
end
```

b)

```
for i = 4 : 4 : 48
    disp(i);
end
```

c)

```
for i = 4 : 48 : 4
    disp(i);
end
```

d)

```
for i = 4 : -4 : 48
    disp(i);
end
```

e) None of the above

2. Which of the following codes prints 5 times "Hello World" to the screen?

a)

```
for i = 0 : 4
    disp("Hello World");
end
```

b)

```
for i = 0 : 5
    disp("Hello World");
end
```

c)

```
for i = 0 : 2 : 5
    disp("Hello World");
end
```

d)

```
for i = 0 : 2 : 4
    disp("Hello World");
end
```

e) None of the above

3. What is the output of the following program?

```
x = 5;
displayed = "";
while x < 10
```

```
    x = x + 1;
    displayed = displayed + x;
end
disp(displayed);
```

- a) 678910
- b) 5678910
- c) 56789
- d) 6789
- e) 67891011

4. What will be the output of the following program?

```
sum = 0;
for i = 0 : 2 : 20
    sum = sum + i;
end
disp(sum);
```

- a) 55
- b) 210
- c) 110
- d) 50
- e) sum

5. What will be the output of the following program?

```
for i = 1 : 3
    displayed = "";
    for j = 1 : i
        displayed = displayed + "*";
    end
    disp(displayed);
end
```

- a) * * *
- * * *
- * * *
- b) * * *
- * *
- *
- c) *
- * *
- * * *
- d) * * * * * * * *
- e) *

*
*
*
*
*
*
*
*

6. Let say you have the function factorial

```
function result = factorial (N)
```

which calculates $N!$. Which of the following code fragments then calculate the following:

$$C = \frac{N!}{M!(N-M)!}$$

a)

```
C = factorial (N, M)
```

b)

```
C = 0;
for i = 1 : N
    C = C + factorial (N, M);
end
```

c)

```
C = factorial (N) / factorial (M) * factorial (N - M);
```

d)

```
C = factorial (N) / ( factorial (M) * factorial (N - M));
```

e)

```
C = 0;
for i = 1 : M
    C = C + factorial (N, i);
end
```

7. Let say you have the function gcd

```
function result = gcd(x, y)
```

which calculates greatest common divisor of x and y . If the least common multiple of x and y can be calculated as

$$lcm(x, y) = xy / gcd(x, y)$$

Which of the following code fragments calculate the least common multiple of x and y ?

a)

```
lcm = xy / gcd(x, y);
```

b)

```
lcm = x * y / gcd(x, y);
```

c)

```
lcm = 0;
for i = 1 : y
    lcm = lcm + gcd(x, i);
end
```

d)

```
lcm = x * y / gcd(x);
```

e)

```
lcm = 0;
for i = 1: x
    lcm = lcm + gcd(i, y);
end
```

8. Suppose you are given a method named **findSum** that reads numbers a , b , and c from the keyboard, and returns the sum of all numbers between $[a, b]$ in increments of c . Which of the following is the correct header for **findSum** method?

- a) **function** findSum(c)
- b) **function** result = findSum(a, b, c)
- c) **function** result = findSum()
- d) **function** findSum()
- e) **function** findSum(a, b, c)

9. Consider the following program:

```
sum = 0;
num = input("Enter number");
for i = 1 : 9
    sum = sum + num / i;
end
disp(sum);
```

How do you convert this program into a method?

a)

```
function result = f(N)
    sum = 0;
    num = input("Enter number");
    for i = 1 : 9
        sum = sum + num / i;
    end
    disp(sum);
end
```

b)

```
function result = f ()
    sum = 0;
    num = input("Enter number");
    for i = 1 : 9
        sum = sum + num / i;
    end
    disp(sum);
end
```

c)

```
function f(N)
    sum = 0;
    for i = 1 : 9
        sum = sum + N / i;
    end
    disp(sum);
end
```

d)

```
function result = f(N)
    sum = 0;
    N = input("Enter number");
    for i = 1 : 9
        sum = sum + num / i;
    end
    disp(sum);
end
```

e)

```
function f(N)
    sum = 0;
    for i = 1 : 9
        sum = sum + num / i;
    end
    disp(sum);
end
```

10. Let say you have the following function defined:

```
function printStar(n)
    displayed = ""
    for i = 1 : n
        displayed = displayed + "*";
    end
    disp(displayed);
end
```

Which of the following code fragments produce the following output?

```
***
**
*
**
***
```

a)

```
printStar (3);
printStar (2);
printStar (1);
printStar (2);
printStar (3);
```

b)

```
printStar (1);
printStar (2);
printStar (3);
printStar (2);
printStar (1);
```

c)

```
printStar (4);
printStar (3);
printStar (2);
printStar (1);
printStar (2);
```

d)

```
printStar (11);
```

e)

```
printStar (3, 3);
```

11. Let say you have the function triangleSum

```
function result = triangleSum(n)
```

which calculates the sum of all integers from 1 to n

$$1 + 2 + \dots + n = \frac{n(n+1)}{2}$$

The sum of cubes of all integers from 1 to n can be calculated as

$$cubes(n) = 1^3 + 2^3 + \dots + n^3 = \left(\frac{n(n+1)}{2} \right)^2$$

Which of the following functions calculates the sum of cubes of all integers from 1 to n ?

a)

```
function result = cubes(n){
    result = triangleSum(n);
end
```

b)

```
function result = cubes(n){
    result = triangleSum(n) * triangleSum(n);
end
```

c)

```
function result = cubes(n){
    total = 0;
    for i = 1 : n
        total = total + triangleSum(i);
    end
    result = total ;
end
```

d)

```
function result = cubes(n){
    total = 0;
    for i = 1 : n
        total = total + triangleSum(i) * triangleSum(i);
    end
    result = total ;
end
```

e)

```
function result = cubes(n){
    total = 0;
    for i = 1 : n
        total = total + triangleSum(i)^3;
    end
    result = total;
end
```

12. Which of the following is the correct way of defining an array?

- a) myList = (1, 2, 3);
- b) myList = ('1', '2', '3');
- c) myList = [1, 2, 3];
- d) myList = 1, 2, 3;
- e) myList = '1', '2', '3';

13. Which of the following is the correct way of defining a matrix of size 3 by 2?

- a) myMatrix = [1, 2, 3, 4, 5, 6];
- b) myMatrix = (1, 2, 3, 4, 5, 6);
- c) myMatrix = [1, 2; 3, 4; 5, 6];
- d) myMatrix = [1, 2, 3; 4, 5, 6];
- e) myMatrix = (1, 2, 3; 4, 5, 6);

14. Which of the following functions creates and returns an array of 100 integers, where each element of the array is 1.

a)

```
function createArray()
    return 1;
end
```

b)

```
function createArray()
    for i = 1 : 100
        array(i) = 1;
    end
end
```

c)

```
function array = createArray()
    for i = 1 : 100
        array(i) = 1;
    end
end
```

d)

```
function return = createArray()
    for i = 1 : 100
        array(i) = 1;
    end
end
```

e)

```
function createArray(array)
    for i = 1 : 100
        array(i) = 1;
    end
end
```

```
end
end
```

15. Which of the following functions takes a parameter of an array, and returns the sum of elements of that array?

a)

```
function sumArray(array)
    sum(array);
end
```

b)

```
function sumArray(array)
    sum = 0;
    for i = 1 : length(array)
        sum = sum + array(i);
    end
end
```

c)

```
function sum = sumArray(array)
    sum = 0;
    for i = 1 : length(array)
        sum = sum + array(i);
    end
end
```

d)

```
function sum = sumArray()
    sum = 0;
    for i = 1 : length(array)
        number = input();
        sum = sum + number;
    end
end
```

e)

```
function sum = sumArray(array)
    sum = 0;
    for i = 1 : length(array)
        number = input();
        sum = sum + number;
    end
end
```

16. Which of the following can be used to get the number of elements of an array?

- a) arr.length
- b) length(arr) - 1
- c) arr.size
- d) length(arr)
- e) arr.size()

17. Which of the following functions takes a parameter of an array and a number N, and returns true if the array contains N and false otherwise?

a)

```
function containsIt(array, N)
    containsIt(array, N);
end
```

b)

```
function containsIt(array, N)
    for i = 1 : length(array)
        if array(i) == N
            result = true;
        else
            result = false;
        end
    end
end
```

c)

```
function containsIt(array, N)
    result = false;
    for i = 1 : length(array)
        if array(i) == N
            result = true;
        end
    end
end
```

d)

```
function result = containsIt(array, N)
    result = false;
    for i = 1 : length(array)
        if array(i) == N
            result = true;
        end
    end
end
```

e)

```
function containsIt(array, N)
    N = input();
    result = false;
    for i = 1 : length(array)
        if array(i) == N
            result = true;
        end
    end
end
```

18. What will be the output of the following program?

```
x = [1, 2, 3];
disp("x(1) = " + x(1));
```

- a) Program gets a compile-time error because the number of elements in the array is not defined.
- b) Program gets a run-time error because there is no initial values for the array elements.
- c) Program runs in a proper way and displays "x(1) = 1".

d) Program gets a run-time error because x(1) is undefined.

e) Program runs in a proper way and displays "x(1) = 2".

19. Assume that we have method which calculates the sum of the numbers between [1,n] with the following signature: function result = findSum(n). What will be the output of the following code?

```
a = [3, 4, 5];
b = findSum(a(1));
c = findSum(a(2));
disp("b=" + b + " c=" + c);
```

- a) b=10 c=15
- b) b=10 c=10
- c) b=15 c=15
- d) b=1 c=2
- e) b=6 c=10

20. Which of the following functions takes a parameter of an array filled with numbers, and returns the largest element of that array?

a)

```
function findGreatest(array)
    for i = 1 : length(array)
        if array(i) > max
            max = array(i);
        end
    end
end
```

b)

```
function findGreatest(array)
    max = 0;
    for i = 1 : length(array)
        if array(i) > max
            max = array(i);
        end
    end
end
```

c)

```
function max = findGreatest(array)
    max = 0;
    for i = 1 : length(array)
        if array(i) > max
            max = array(i);
        end
    end
end
```

d)

```
function max = findGreatest(array)
    for i = 1 : length(array)
        if array(i) > max
            max = array(i);
        end
    end
```

```

    end
end

```

e)

```

function max = findGreatest(array)
    for i = 0 : length(array)
        if array(i) > max
            max = array(i);
        end
    end
end
end

```

21. What will be the output of the following program?

```

x = [1, 2, 3; 4, 5, 6];
disp("x(2, 1) = " + x(2, 1));

```

- a) x(2, 1) = 3
- b) x(2, 1) = 5
- c) x(2, 1) = 4
- d) x(2, 1) = 1
- e) x(2, 1) = 6

22. Given T as an integer, how do you check if T is not between 0 and 11?

- a) if $T < 0 \parallel T > 11$
- b) if $T < 0 \&\& T > 11$
- c) if $0 < T < 11$
- d) if $T > 0 \parallel T < 11$
- e) if $T > 0 \&\& T < 11$

23. What is the value of integer variable x after the execution of the following line?

```

x = 1 / 2 + 1 / 4 + 1 / 4 + 1 / 2;

```

- a) 0
- b) 0.5
- c) 1
- d) 1.5
- e) 2

24. What is the output of the following program?

```

a = 4;
b = 8;
c = 12;
d = 8;
disp((a + b) * c + d - a * c);

```

- a) 160
- b) 152
- c) 240
- d) 60
- e) 104

25. Suppose x is 67, what is the output of the following code?

```

if x < 20
    disp("1");
else
    if x < 30
        disp("2");
    else
        if x < 50
            disp("3");
        else
            if x < 80
                disp("4");
            else
                disp("5");
            end
        end
    end
end
end
end

```

- a) 1
- b) 2
- c) 3
- d) 4
- e) 5